CELESTIAL MECHANICS × AI · 2026

From the next token to the next theorem

How a model works — and how we should use it

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Draft v0.1

45 min

What the model sees: ten integers

EN · ONE SENTENCE

Celestial mechanics is the poetry of orbits.

Cel	estial	_mechanics	_is	_the	_poetry	_of	_or	bits	.
69126	90010	52347	382	290	38153	328	503	24043	13

46 characters → **10** tokens → 10 integers · _ marks a leading space

CHINESE · FOUNDATIONS OF CELESTIAL MECHANICS

本书系统地介绍了天体力学

本	书	系统	地	介绍	了	天	体	力	学
4087	20121	31936	5557	37887	4531	867	7074	11343	8590

12 characters → **10** tokens → 10 integers · two-character words often map to one token

“This book systematically introduces celestial mechanics.”

01 Where do these integers come from?

02 How does it predict the next one?

03 How do we put it to work?

Put the model back in the system



Inner ring: one call · Middle ring: across calls · Outer ring: engineering boundary

01	**LLM**	A probability generator: tokens in, a distribution over the next token out
02	**Prompt**	This call's conditions: system / user / few-shot
03	**Context**	Every token visible to this call; the window is finite
04	**Memory**	State beyond the window: write / retrieve / compress / forget
05	**Agent loop**	Observe → act → observe; state lives outside
06	**Harness**	Tools, sandbox, permissions, verification, logs

> The model is the CPU; prompt, context, memory, loop and harness are the runtime.

Acceleration: from basic use to gold medals

2023 · BEFORE

Basic use

- Explain concepts, search the literature, write code
- Do simple calculations, often wrong
- Complex proofs still need people
- Role: assistant

→ 19 months

2025 · AFTER

Research level

- IMO 2025 gold level 35/42
- Lean proofs, machine-checkable
- OpenAI 2026: 372 problems, ≈3 h each (report)
- Role: candidate generator

2024.07 **28/42** silver → 2025.07 **35/42** gold level — 12 months, +7 points

> This is a staircase with **three steps**.

Timeline: three steps

2023–24

Tools: search + verification

2023.12
FunSearch
LLM + evaluator + evolutionary search; a new cap-set construction

2024.01
AlphaGeometry
Neuro-symbolic; IMO geometry 25/30

2024.07
AlphaProof
Lean + RL; IMO 2024 silver 28/42 (+AG2)

2024–25

Reasoning: RL + test-time compute

2024.09
OpenAI o1
RL teaches thinking; test-time compute

2025.01
DeepSeek-R1
RLVR; reasoning behavior emerges; open weights

2025.05
AlphaEvolve
Evolutionary coding agent; 4×4 complex matrix product in 48 multiplications

2025–26

Scale: from benchmarks to open problems

2025.07
IMO 2025
Gold level 35/42; natural language, no tools

2025
Formal gold (claimed)
Aristotle et al.; Lean-checkable

2026.05–10
OpenAI
Unit-distance counterexample; NS / Riemann-related (claimed); 372 problems (report)

> Every leap combines a **generator, a verifier and search**.

The motor and the assembly line

A story Eric Schmidt tells — and the AI parallel.

When the electric motor was invented, factory owners did the obvious thing: remove the central steam engine, put a motor in its place, and change nothing else. Efficiency rose by about 30%.

Decades later, someone realized the motor could be small. Every machine could have its own. The factory could be designed around the flow of production itself; the power source no longer dictated the layout.

The assembly line was born. Efficiency rose by an order of magnitude.

30%

same layout
The motor takes the steam engine's place

10×+

new layout
Small motors per machine; the factory is rebuilt around flow

The gap is the **redesign of work**.

> AI today: "ask AI" inside the old step → 30%. Redesign the workflow around agents, tools, context, verification, and a loop → 10×+.

> The gain is in the **layout**.

Two attempts

Same intention. Different loop.

OPENCLAW · EARLY

Hands off

- I let the agent write a paper.
- The model was weak; errors multiplied.
- I stepped back too far.

Result: no paper.

CODEX · GPT-5.3–5.5 · ISTART

Human in the loop

- A continuation of my postdoc work; no time to write or think deeply.
- I set the goal; the model drafted the main content.
- It made errors: e.g. claimed $Ax = b,\ b \neq 0 \to x = 0$. I corrected it.
- After 3 rejections, GPT-6.1 Sol found more issues; I fixed them.

Result: progress, with checks.

> The difference was the **loop**.

Low-hanging fruit

What the two attempts taught me.

01 · HUMAN

In the loop

Full hands-off fails. The human sets the goal, corrects errors, and decides when to stop.

02 · MODEL

Quality + checks

A better model gives a real gain—and still makes mistakes and gets stuck. Check.

03 · PROBLEM

Choose well

Writing the paper is the easy part. Choosing the problem is the hard part.

When OpenAI began mass-producing open-problem results, I stopped writing marginal papers. The old conjectures were written for humans; machines are a different audience.

> Which problem is **worth solving**?

A young person's opportunity

Celestial mechanics is becoming a data field. The scarce skill is finding the signal.



Data

Gaia. Satellite tracking. Surveys. N-body simulations.

`More data than hands.`



Craft

Denoise. Model. Verify. Agents run the pipeline.

`The new leverage.`



Signal

Resonances. Anomalies. New orbits.

`The old questions.`

More data than hands. New tools. Old questions. **That is a young person's advantage.**

> The bottleneck is **signal**.

From the next token to the next theorem

`token` → `probability` → `candidate` → `verification` → **`theorem`**

Probability gives breadth

Verification gives correctness

People set the direction

> The next theorem still begins with the **next token**.